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AI Development Guide

OWNED & SUPPORTED BY

NiceGUI Base

Support

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VERSION

NiceGUI Base 3.0.0a8

NiceGUI 3.15.0

REFERENCE

Start Here

Start from the engineering intent, visually compare the governed authorities, then open only the reference you need.

Search ⌘K / Ctrl+K

Theme · System

Appearance

NICEGUI BASE · DEPARTMENT STANDARD

What are you building?

Start from intent. NiceGUI Base will lead you to the governed pattern, layout, visualization, component, recipe and scaffold instead of making you browse hundreds of references one by one.

10
Patterns

58
Analytics

34
Components

8
Recipes

Choose an intent

See the recommended pattern, recipe, scaffold, and related governed references before leaving this page.

MONITOR

Monitor a process

Health, trends, limits, alerts and review queues.

Recurring health and exception review

INVESTIGATE

Investigate an excursion

Affected/control evidence, commonality and drill-down.

Evidence-led root-cause investigation

COMPARE

Compare tools or populations

Aligned distributions, deltas and supporting records.

Side-by-side population or tool decisions

EXPLORE

Explore engineering data

Filter, inspect, chart and drill into arbitrary records.

Filter-first record exploration

DASHBOARD

Build an operational dashboard

KPI hierarchy, trends and exception-focused detail.

KPI, trend and exception overview

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REFERENCE

Developer Scaffolding

This saved-link route now points developers to deterministic pattern and recipe starters.

Search⌘K / Ctrl+K

Theme · System

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Compatibility authoring surface

Manual composition is retired from the normal reference journey. Start from a canonical pattern or recipe with the installed CLI; domain logic stays in your application services.

CLI · CREATE-PATTERN

Choose a pattern

Use a registered page hierarchy and responsive slot contract.

CLI · CREATE-RECIPE

Choose a recipe

Use a bounded semiconductor workflow and its data contract.

AGENT WORKFLOW

Read the agent path

Resolve a requirement into an authority before coding.

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Design System

The deterministic visual contract used by the reference explorer, framework components, and generated applications.

Search

⌘K / Ctrl+K

Theme · System

Appearance

Design System Authority

See the token in use first; the compact contract beside each specimen is the source for Python consumers, components, CSS, and generated applications.

Import from nicegui_base.design; choose semantic roles before extending the framework.

Spacing scale

Use for recurring distances; prefer semantic gaps for composition.

LIVE USE

Tight

--cui-space-1 · 4px

Related

--cui-space-2 · 8px

Stack

--cui-space-4 · 16px

Content

--cui-space-6 · 24px

Section

--cui-space-10 · 40px

Page

--cui-space-16 · 64px

classes("cui-stack cui-gap-content")

Copy semantic API

RECOMMENDED USE

Semantic gaps

Use for page, section, content, stack, cluster, and control relationships.

LIVE USE

Page

Section

Content

Control

page_section(..., gap="section")

Copy semantic API

RECOMMENDED USE

Use for page, section, content, stack, cluster, and control relationships.

Don't: invent a local value when a token exists.

<

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REFERENCE

Settings

Manage appearance and Explorer identity without leaving the department reference shell.

Search

⌘K / Ctrl+K



Theme · System

Appearance

Appearance

These preferences apply to the Explorer and its mounted reference previews. System follows the operating-system color scheme.

Theme

System

Light

Dark

Density

Comfort

Compact

Dense

Motion

Normal

Reduced

Account

Identity is read from the company environment and is display-only here.

AccessKey

Company identity value

Member

Role

Reference Explorer display value

Member

Department

Reference Explorer display value

Department unavailable

About this reference

Use the canonical registries, patterns, tables, visualizations and recipes shown here in application code.

NiceGUI Base 3.0.0a8

NiceGUI 3.15.0

Reference Explorer

<

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NiceGUI Base 3.0.0a8

NiceGUI 3.15.0

REFERENCE

Design System

One discoverable inventory over canonical NiceGUI Base registries, grouped by the job each capability solves.

Search

⌘K / Ctrl+K



Theme · System

Appearance

Search catalog

Refine results · type, family, intent, data, domain, relation

Analytics Studio

Name, tag, intent, data shape, dc

465 canonical Reference Explorer entries are searchable. Choose a family or search to inspect individual capabilities.

Shell/Layout

1 canonical entry

Examples: Layout

Browse family

Controls

35 canonical entries

Examples: ButtonGroup, SplitButton, Divider

Browse family

Forms/Overlays

25 canonical entries

Examples: Form Filter Overlay, Form, FilterBar

Browse family

Tables

7 canonical entries

Examples: Table, DataTable, ServerDataTable

Browse family

Content/Workflow

25 canonical entries

Examples: Content, Jobs, MetricCard

Browse family

Generic Visualizations

25 canonical entries

Examples: Visualization, AreaChart, BarChart

Browse family

Engineering Components

Generic Application Patterns

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REFERENCE

Components

Scan governed component specimens first; open a detail only for deeper states, configuration, accessibility, or code.

Search

⌘K / Ctrl+K

Theme · System

Appearance

ButtonGroup

Interactive specimen · Use the real control below; interactions are recorded in Event / debug output.

Run

Pause

Reset

ACTIONS · LIVE

ButtonGroup

Group related actions without visual fragmentation

Best for · toolbar actions · mode actions

→ Open example

☆ Favorite

Compare

SplitButton

Interactive specimen · Use the real control below; interactions are recorded in Event / debug output.

Export Data

ACTIONS · LIVE

SplitButton

Primary action with closely related alternatives

Best for · export alternatives · run options

→ Open example

☆ Favorite

Compare

Divider

Interactive specimen · Use the real control below; interactions are recorded in Event / debug output.

Measurement summary

Run details

SURFACES · LIVE

Divider

Subtle structural separation

Best for · sections · menus

→ Open example

☆ Favorite

Compare

CollapsiblePanel

Interactive specimen · Use the real control below; interactions are recorded in Event / debug output.

Advanced settings

Secondary configuration steps in context.

Accordion

Interactive specimen · Use the real control below; interactions are recorded in Event / debug output.

Sampling

Every 10 values

Notification

Chip

Interactive specimen · Use the real control below; interactions are recorded in Event / debug output.

SEARCH

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Application Patterns

Choose a finished application structure, see the real governed composition, then adapt business logic in your own services.

Search

⌘K / Ctrl+K

Theme · System

Appearance

LIVE GOVERNED COMPOSITION

Dashboard

kpis · trend · exceptions

Dashboard

At-a-glance process performance with trends and exceptions.

Lots today

128

↑ +6%

Across three qualified tools

In control

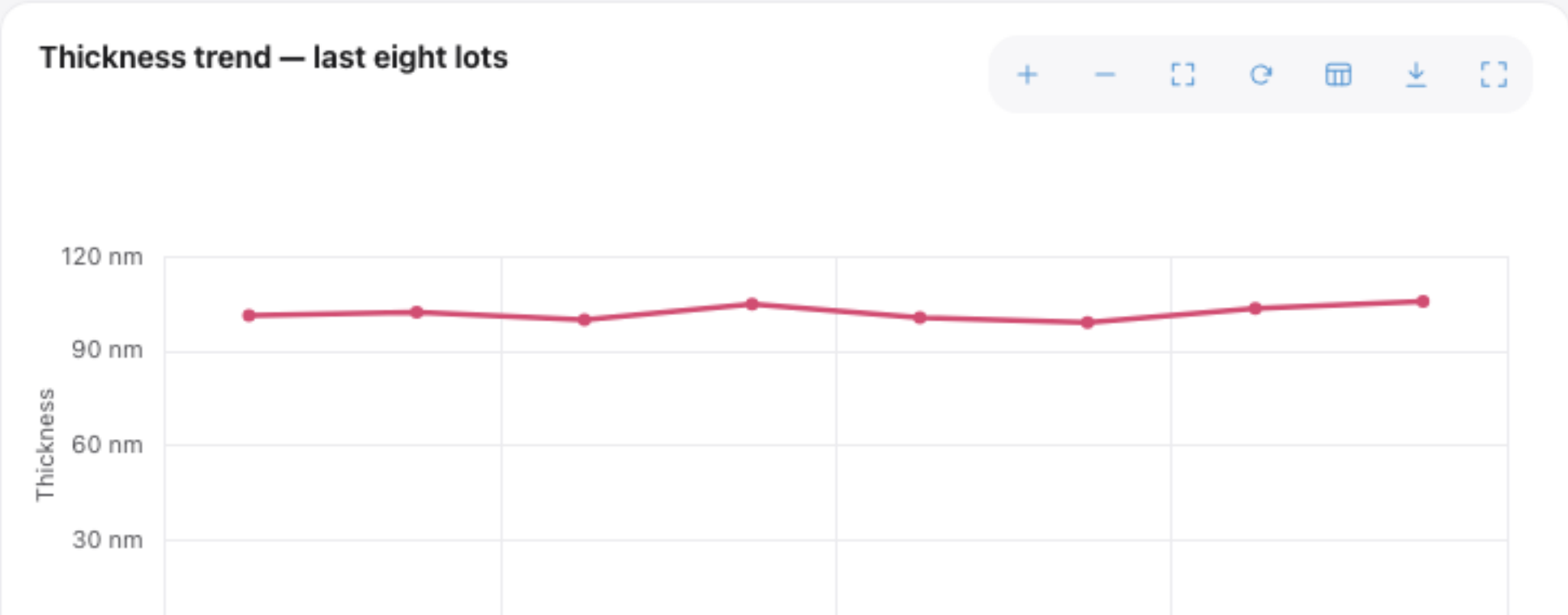
96.9%

↑ +0.8 pp

Open exceptions

3

Two require engineer review



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Visualizations

See what all 58 governed analytical surfaces actually look like before opening a full interactive reference.

Search

⌘K / Ctrl+K



Theme · System

Appearance



SPC · LIVE

SPC I-MR

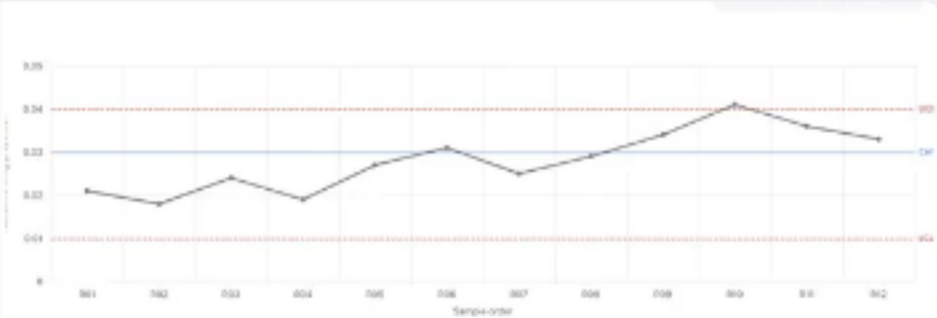
Individuals and moving-range control chart

Best for · continuous measurements without rational subgroups

→ Open example

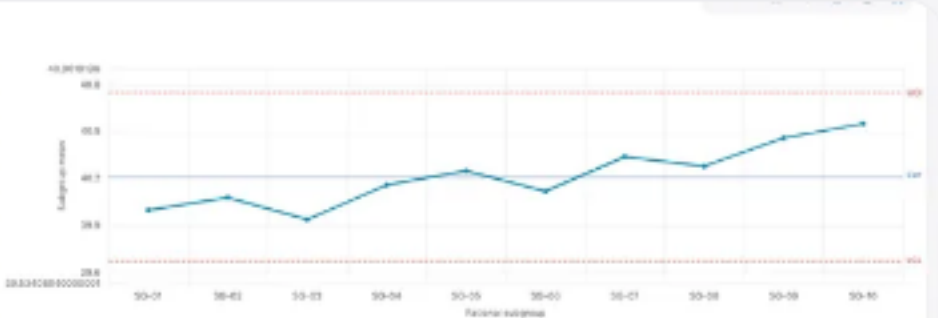
☆ Favorite

Compare



ANALYTIC · SPC

SPC P



SPC · LIVE

SPC Xbar-R

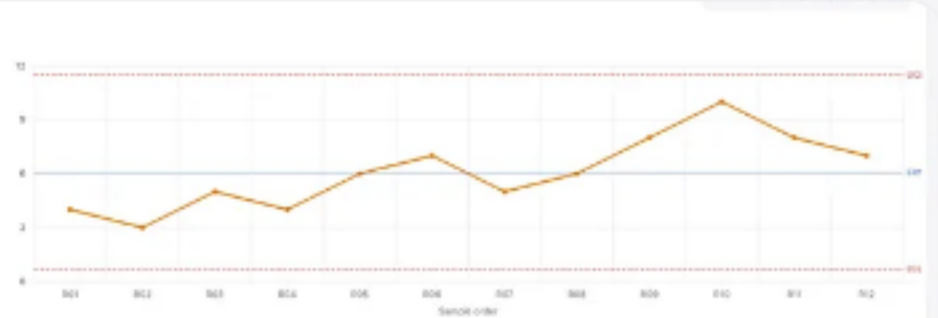
Xbar-R subgroup control chart

Best for · small equal-size rational subgroups

→ Open example

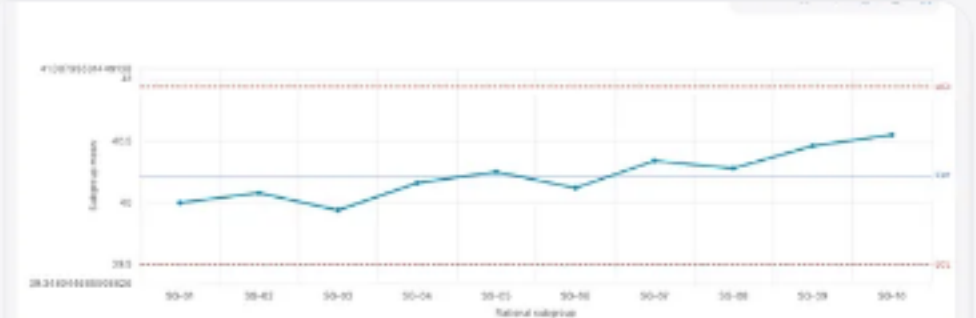
☆ Favorite

Compare



ANALYTIC · SPC

SPC Np



SPC · LIVE

SPC Xbar-S

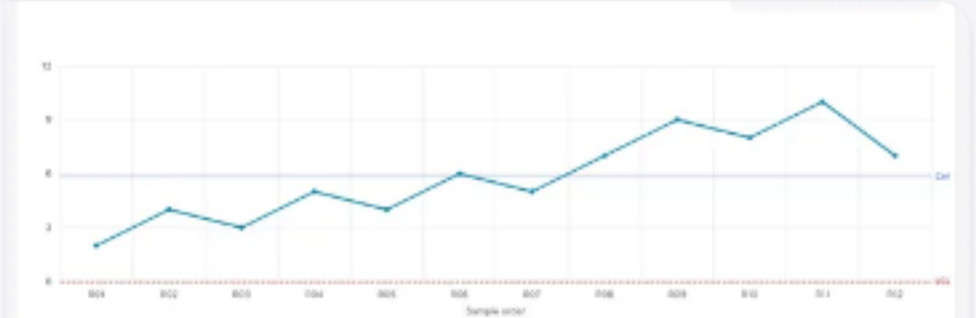
Xbar-S subgroup control chart

Best for · equal-size subgroup variability matters

→ Open example

☆ Favorite

Compare



ANALYTIC · SPC

SPC C

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Semiconductor Recipes

Scan the domain workflows visually, compare their contracts, then open the recipe that matches the engineering question.

Search

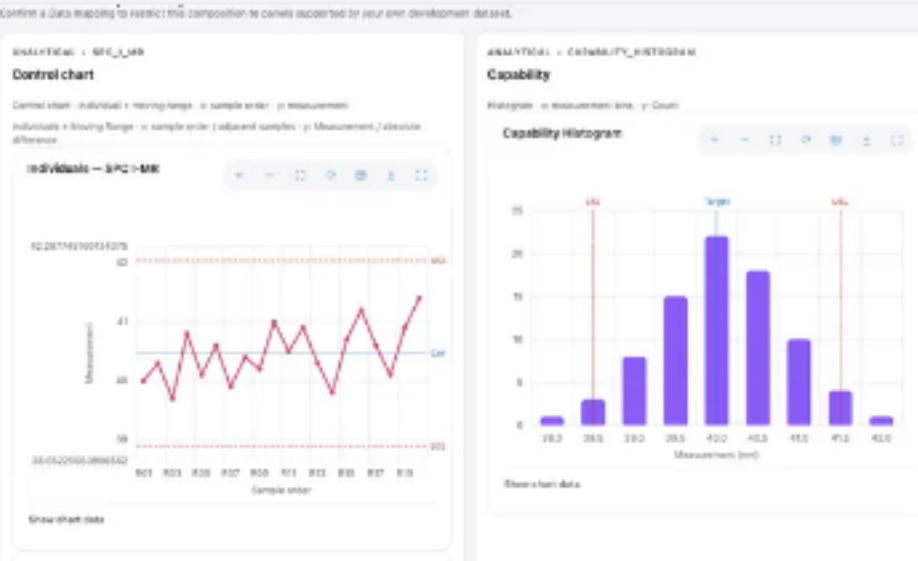
⌘K / Ctrl+K



Theme · System



Appearance



SEMICONDUCTOR RECIPE · LIVE

SPCMonitor

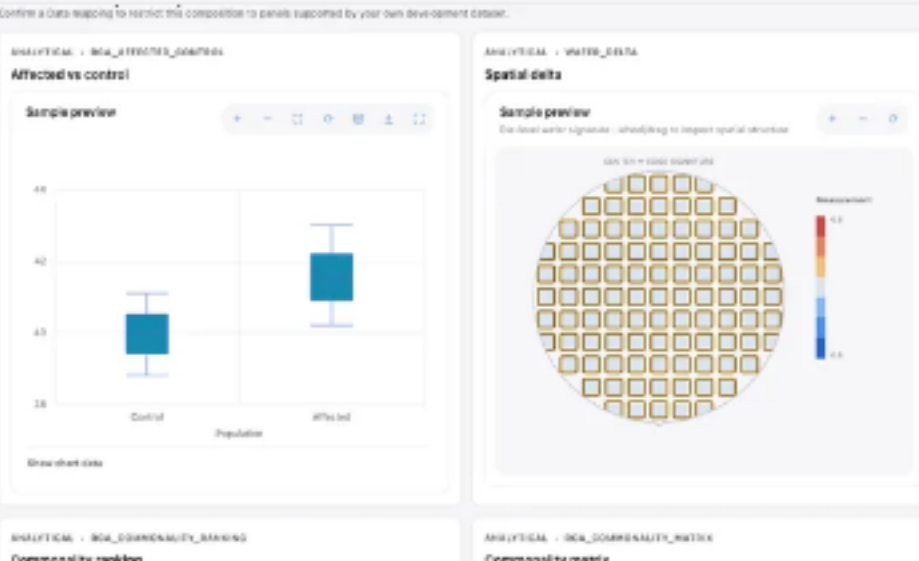
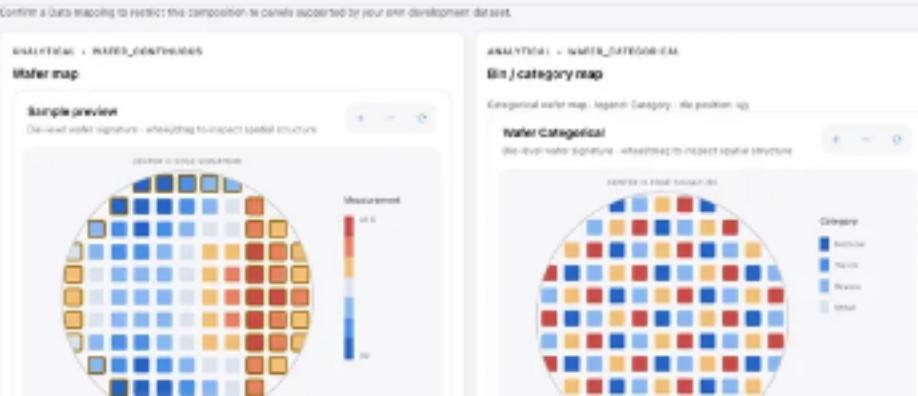
Monitor process stability, capability and violations with drill-down to manufacturing context.

Best for · monitor a process parameter · detect process drift or control-rule violations

→ Open example

☆ Favorite

Compare



SEMICONDUCTOR RECIPE · LIVE

ExcursionDefenseLine

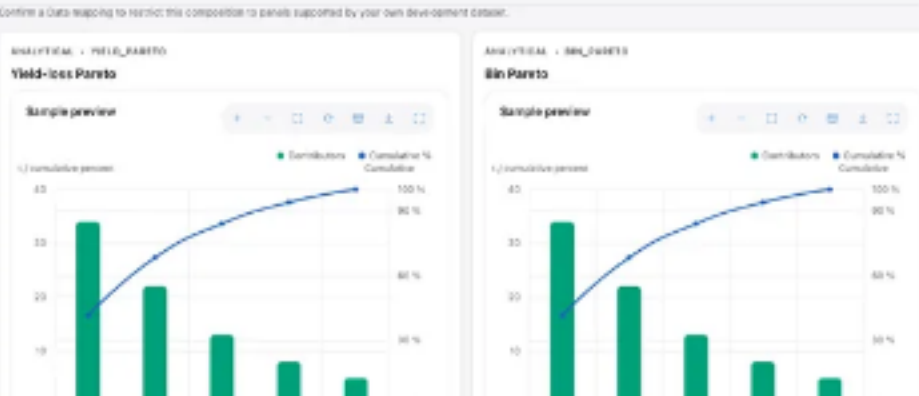
Triage an excursion with affected/control populations, spatial localization, commonality and genealogy evidence.

Best for · build an excursion defense line · triage affected versus control material

→ Open example

☆ Favorite

Compare



SEMICONDUCTOR RECIPE · LIVE

FdcToolHealth

Diagnose equipment health from aligned sensor traces, envelopes, events and multivariate fingerprints.

Best for · analyze FDC traces · check tool or chamber health

→ Open example

☆ Favorite

Compare



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Data & Tables

A governed production reference for tables, record management, data exploration and provider-scale grids.

Search ⌘K / Ctrl+K

Theme · System

Appearance

Production table lab

The governed production reference for tables, record management, data exploration and provider-scale grids. One active lab keeps the page fast; every lab reuses the same semiconductor data contract.

- Grid
- Actions
- Edit
- Visualize
- Import
- Detail
- Server
- States

Grid

The production read-oriented grid: query, layout, views and persisted state.

64 records

Persisted view

- Default
- Watch / OOS
- Unreviewed
- Clear filters
- Clear sorting
- Reset view

64

Records

49.820 nm

Mean measurement

94.1%

Mean yield

8

OOS

View: default · 64 records

Lot and wafer monitoring

Search, filter, sort and save the view you want to reuse.

Search records /

Columns

Compact

Export

Refresh

☐	Record	Timestamp	Lot	Waf...	Tool	Cha...	Operation	Recipe	Measurem...	Targ	Actions
--------------------------	--------	-----------	-----	--------	------	--------	-----------	--------	-------------	------	---------

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NiceGUI 3.15.0

REFERENCE

Diagnostics

Understand runtime health, framework coverage, and the few actions that need attention.

Search⌘K / Ctrl+K

Theme · System

Appearance

System health

A quick answer to whether the installed reference is ready to use.

Reference Explorer

Healthy

Candidate NGB-20260907-G2.6 · NiceGUI Base 3.0.0a8 · NiceGUI 3.15.0

Generic application patterns

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Live examples10 / 10

Semiconductor analytics

Available58

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Semiconductor recipes

Available8

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Live examples8 / 8

Catalog family coverage

Every capability family must be backed by at least one discoverable canonical authority entry.

shell/layout

1 discoverable entry

controls

35 discoverable entries

forms/overlays

tables

<

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AI Development Guide

A compact, task-first guide for humans and AI agents using the installed NiceGUI Base APIs.

Search

⌘K / Ctrl+K

Theme · System

Appearance

Requirement → runnable reference

The same deterministic sequence is readable by a person and discoverable by an agent.

1 · REQUIREMENT

Investigate chamber drift and compare affected/control distributions

2 · RECOMMENDATION

Pattern: analysis_workspace · Recipe: excursion-defense-line

3 · AUTHORITIES

analytics:fdc_multi_sensor · analytics:rca_affected_control · component:data_table

4 · SCAFFOLD

nicegui-base create-recipe ./chamber-drift --name "Chamber Drift Review" --recipe excursion-defense-line

5 · BUSINESS/DOMAIN LOGIC

app/services and app/repositories; keep provider queries and domain calculations out of page callbacks

6 · EXTENSION RULE

Prefer registered pattern/component/visualization and token APIs; isolate a documented escape hatch only when no authority fits

7 · EXPECTED RESULT

A separately runnable app with governed filters, linked chart/table state, accessible empty/error states, and domain logic isolated in app-owned services

```
nicegui-base agent-context "Investigate chamber drift and compare affected/control distributions"
nicegui-base recommend-pattern "investigate chamber drift"
nicegui-base recommend-visualization "compare affected control distributions"
```





<

START

Overview

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Engineering & RCA

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NiceGUI Base / Metrology Enginee...

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LOCAL

Foundation

Inspect the exact visual primitives from which every application is constructed.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Semantic color system

These are design roles, not a decorative palette. Apps should never invent independent visual colors.

Page
--cui-page

Primary surface
--cui-surface

Secondary surface
--cui-surface-secondary

Elevated
--cui-surface-elevated

Accent
--cui-accent

Success
--cui-success

Warning
--cui-warning

Danger
--cui-danger

Info
--cui-info

Primary text
--cui-text-primary

Secondary text
--cui-text-secondary

Border
--cui-border-default

Typography

Dense enterprise readability with restrained hierarchy and tabular numerics where values must scan precisely.

Type scale

Display — Strategic overview

Page — Equipment Health

Heading — Affected population



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Shell Primitives

One responsive navigation state machine: expanded desktop, icon-only desktop, or temporary mobile overlay.

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Resize the real browser window for responsive proof

Responsive navigation contract

Desktop has one collapse control inside the rail. Below 900px the rail is removed from interaction and one navigation trigger appears with the header actions — never beside the application title.

Expanded desktop

Icon-only desktop

Mobile overlay

Resize the browser to exercise the states. There is intentionally no standalone mobile-navigation demo action because that created a competing navigation system.

Header interaction contract

Application settings and user profile are real structured popovers. Settings navigates to the canonical settings workspace; the user control exposes identity and preferences instead of firing a toast-only action.



Structural acceptance rule

Exactly one navigation affordance is interactive per viewport. Sidebar/footer compression, duplicate menu triggers, content underlap, or lost navigation after resize are release blockers.



- START
- Overview
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- ELEMENTS
- Controls

- Forms & Overlays
- DataTable Lab
- Charts
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Control Laboratory

Every primitive shown against normal, disabled, selected, error, long-content and compact conditions.

THEME

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Dark

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Reduced

Resize the real browser window for responsive proof

Buttons and action hierarchy

Intent hierarchy

Primary

Secondary

Tertiary

Ghost

Danger

Save changes

Processing

Disabled

Icon, group and split actions

Search

Refresh

More

Delete

Day

Week

Month

Export

More

Navigation controls

Tabs, segmented controls and user menus are especially important stock-Quasar leakage surfaces.

Tabs

Summary

Evidence

History

Disabled

Summary panel content

Segmented + user menu

Trend

Table

Split

PE

About this lab

png_sha256=d5c0dd74cb398262325e837f1fb0875c4f9860dec7fb118ad22c98187ee1b6f1



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LOCAL

Forms, Overlays & Feedback

Exercise real form anatomy, dirty state, dialogs, drawers, menus, progress, alerts and failure recovery.

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Resize the real browser window for responsive proof

Production form

Routing

Changes here are intentionally marked dirty for navigation protection.

Preferred tool

ETCH-021

×

▼

Recipe ^{*}

ETCH_R18

Automatic dispatch

Advanced limits

^

Collapsible form-section normalization.

Warning threshold

41.5

nm

Stop threshold

42.5

nm

<

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Enterprise DataTable Gallery

Five distinct production patterns prove different table jobs instead of repeating one measurement dataset.

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Resize the real browser window for responsive proof

1 • Measurement population

High-volume read-only analysis: search, sort, filter, resize, select, conditional values, status, sparklines, bulk actions and row inspection.

Measurement population

320 deterministic records · dense analytical browsing · double-click any row for inspection

Search records

/

Columns

Compact

Export

Refresh

☐	Measurement	Lot	Wafer	Tool	Parameter	Actions
☐	MEAS-100000	L260100	W01	ETCH-014	CD	Actions
☐	MEAS-100001	L260100	W02	ETCH-021	Thickness	Actions
☐	MEAS-100002	L260100	W03	CVD-008	Overlay	Actions
☐	MEAS-100003	L260100	W04	CMP-004	Etch Rate	Actions
☐	MEAS-100004	L260100	W05	PVD-011	CD	Actions
☐	MEAS-100005	L260100	W06	MET-030	Thickness	Actions
☐	MEAS-100006	L260100	W07	ETCH-014	Overlay	Actions
☐	MEAS-100007	L260100	W08	ETCH-021	Etch Rate	Actions
☐	MEAS-100008	L260101	W09	CVD-008	CD	Actions
☐	MEAS-100009	L260101	W10	CMP-004	Thickness	Actions
☐	MEAS-100010	L260101	W11	ETCH-014	Thickness	Actions



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Visualization Recipe Gallery

Compact reusable recipes demonstrate analytical intent, markings, statistics, interactions and specialist visual forms—not just renderer types.

THEME

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Resize the real browser window for responsive proof

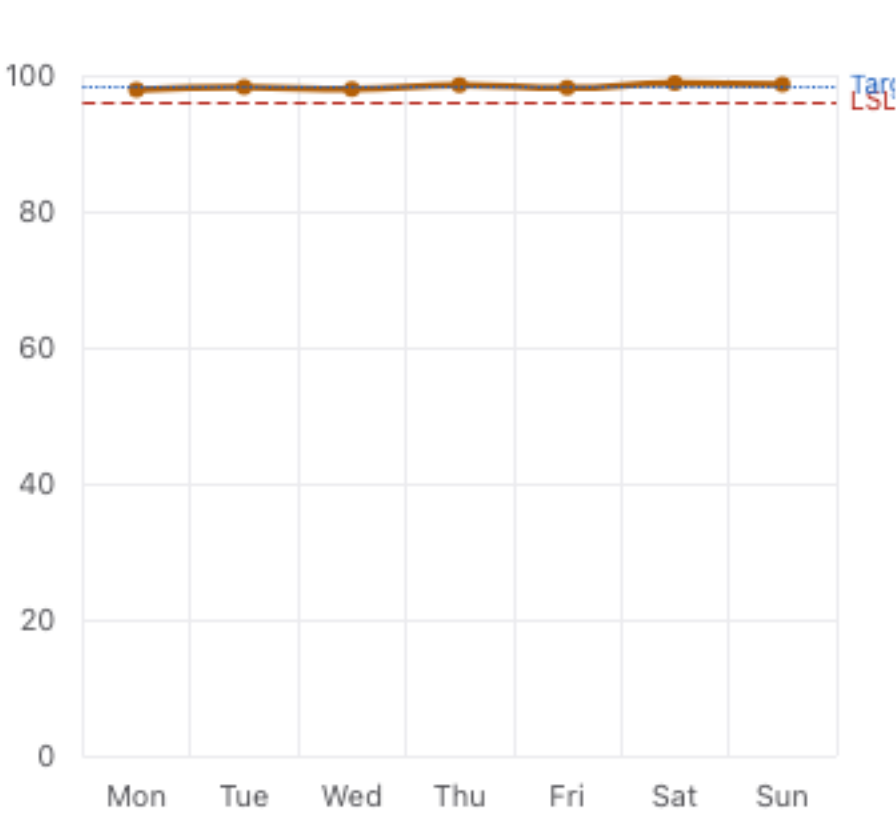
Trend & time recipes

Premade temporal patterns include limits, target context, multi-series comparison, events, confidence bands and step changes.

Yield trend + target

+

−

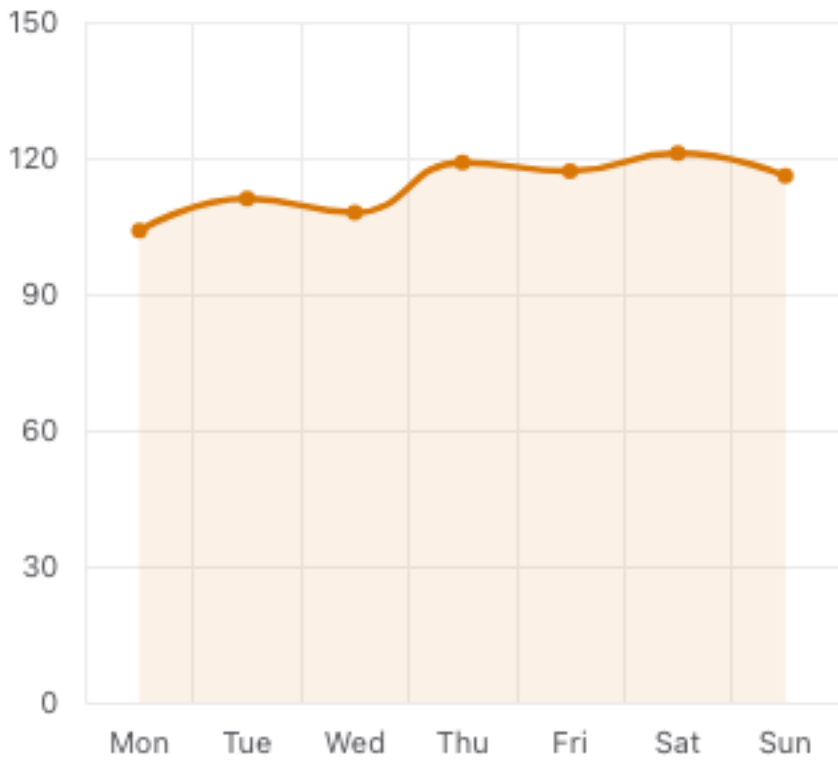


Show chart data

Throughput area

+

−

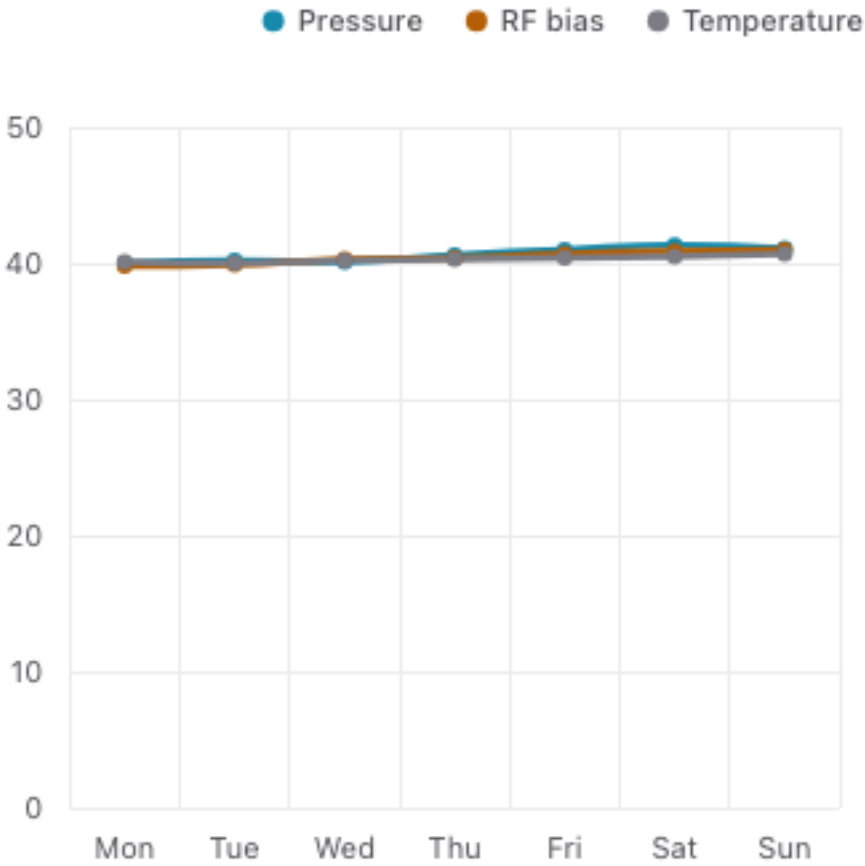


Show chart data

Multi-sensor overlay

+

−



Show chart data

Step change + PM marker

Confidence-band trend



<

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LOCAL

Content, Workflow & Commands

Viewers and workflow components should visually inherit NiceGUI Base rather than their underlying third-party defaults.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Entity and property presentation

ETCH-021 / CH-3

Watch

Critical etch chamber investigation

Chamber Area: ETCH Recipe: ETCH_R18

Key/value

Lot

L260142

Wafer

W12

Owner

Process Engineering

Property grid

STATUS

Watch

LAST PM

17 days

PRESSURE

12.8 mTorr

Comparison metric

Mean CD

42.18 nm

40.06 nm

+2.12

Delta and trend indicators

Standalone metric indicators preserve non-color directional semantics.

↑ +2.12 nm

↓ -1.8%

— Stable

Hierarchy and viewers

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- Charts
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- Engineering & RCA

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Engineering & RCA Laboratory

Purpose-built investigation cockpit: scope, process behavior, population separation, evidence, commonality, contradictions and timeline.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Affected population

84 wafers

12 lots

Matched controls

215 wafers

exact route strategy

Evidence confidence

High

2 strong channels

Contradictions

1

retained in decision

INVESTIGATION	OWNER	STAGE	UPDATED
EXC-1042 · Chamber 3 process degradation	Process Engineering	Evidence review	2 minutes ago

Investigation identity and specification context

ETCH-021 · Chamber 3

Etch critical dimension

Watch

TOOL / CHAMBER

ETCH-021 / CH-3

RECIPE

ETCH_R18

LAST PM

17 days

LOTS AFFECTED

12 lots

Specification semantics

Normal

Watch

Critical

40.100 nm · In spec

41.950 nm · Near upper limit

43.100 nm · Above USL

43.100 nm · Above USL

Current

42.180 nm

Baseline

40.060 nm

Delta

+2.120 nm

High



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NiceGUI Base 3.0.0a8

NiceGUI 3.15.0

LOCAL

State & Failure Laboratory

Every unhappy path must still look intentional and preserve clear recovery behavior.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Canonical application states

No data yet

Connect a source or adjust the time window.

Configure

No matching results

Filters excluded all 18,402 measurements.

Clear filters

Unable to load this content

The synthetic service returned a controlled failure.

Error ID: ERR-LAB-2048

Retry

Connection unavailable

The connection is unavailable. Existing content should remain understandable.

Retry

Access restricted

You do not have permission to view this investigation.

Page not found

The requested synthetic investigation does not exist.

Go back

Generic state view



<

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LOCAL

Performance Laboratory

Interactive workloads expose generation, render assembly, cancellation and query timing instead of hiding work behind a button.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Interactive workload benchmark

Every action updates visible state immediately. The large workload remains opt-in so ordinary lab navigation stays fast.

Local grid workload

Generate deterministic synthetic rows off the UI event loop, mount the real Company DataTable, then expose timing and status.

Idle

0
Rows

—
Generation

—
Table mount

—
Filter benchmark



No stress workload loaded

Choose 10k first, then 100k when you want the full local stress path.



Load 10k rows



Load 100k rows

Cancel / clear

Background-work presentation

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NiceGUI 3.15.0

REFERENCE APP PREVIEW · DASHBOARD

Dashboard

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Yield Performance

Executive-to-engineering overview with one clear reading order

Area

All areas



24 h

7 d

30 d

 Yield

98.4%

↑ +0.7% WoW

 Excursions

12

↑ 3 critical

 Fleet health

94.6%

— stable

 Stale signals

4

⌚ >30 min

Yield trend



100

80

60

40

Excursio
n state



 Resolved

 Monitoring

 Open

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NiceGUI Base 3.0.0a8

NiceGUI 3.15.0

REFERENCE APP PREVIEW · DATA EXPLORER

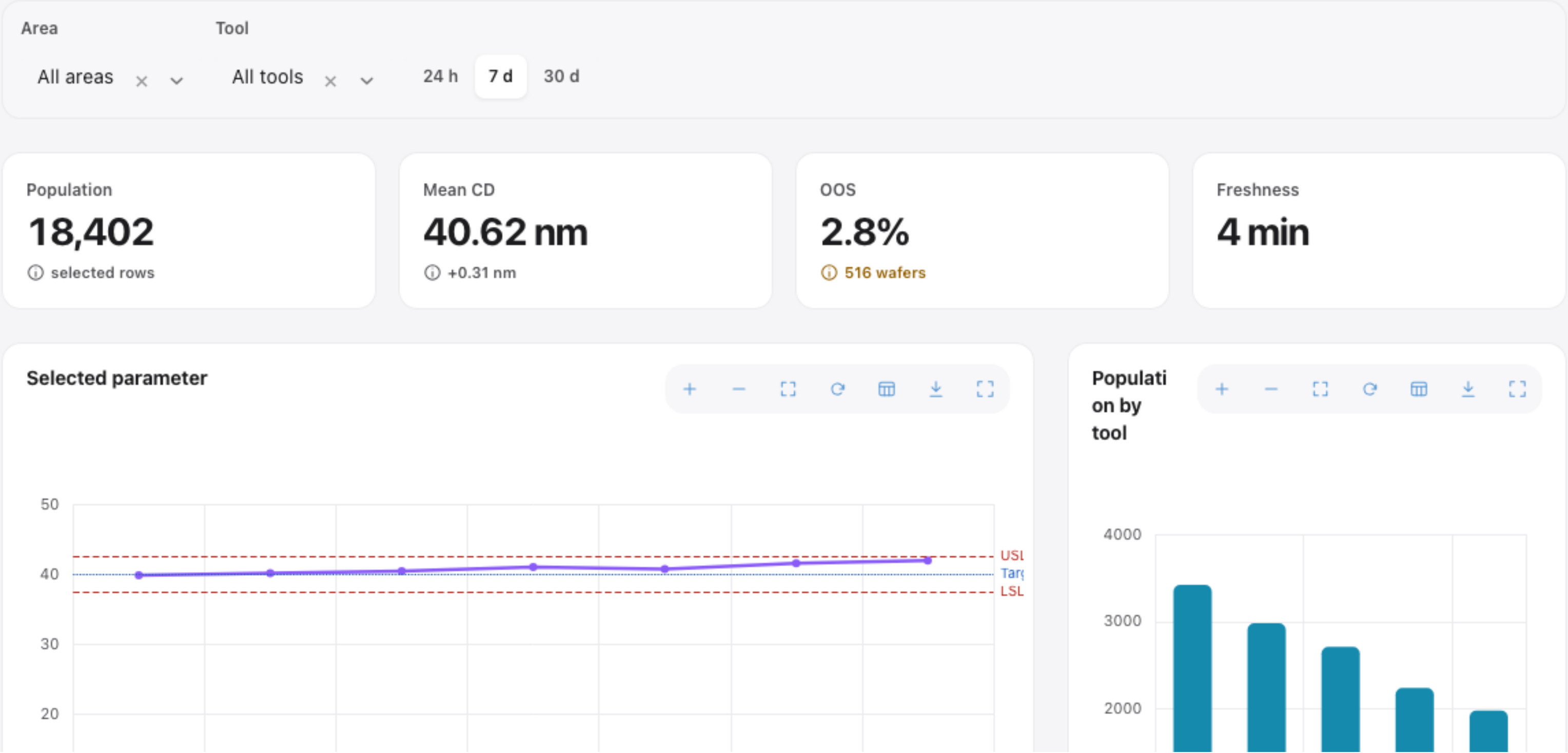
Data Explorer

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Process Measurement Explorer

Filter → summarize → analyze → inspect without losing population context



<

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REFERENCE APP PREVIEW · MASTER DETAIL

Master Detail

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Equipment Fleet

Browse a dense fleet while selected-equipment context stays visible

Area

Tool

All areas × ▾

All tools × ▾

Equipment signals

🔍 Search records

/

📄 Columns

☰ Compact

⬇️ Export

🔄 Refresh

	Measurement	Lot	Wafer
☐	MEAS-100000	L260100	W01
☐	MEAS-100001	L260100	W02
☐	MEAS-100002	L260100	W03
☐	MEAS-100003	L260100	W04
☐	MEAS-100004	L260100	W05
☐	MEAS-100005	L260100	W06
☐	MEAS-100006	L260100	W07
☐	MEAS-100007	L260100	W08

ETCH-021 / CH-3

Watch

Etch chamber

RECIPE

ETCH_R18

LAST PM

17 days

HEALTH

82%

Recent CD

+ − 📐 ↺ 📅 ⬇️ 📏

50

<

START

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REFERENCE APP PREVIEW · CRUD

Crud

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Saved Investigation Views

Create, inspect and manage reusable engineering configurations

Search saved views

All

Mine

Shared

+ New saved view

Saved views

Q Search records

/

Columns

Compact

Export

Refresh

ID	Name	Owner	Status
☐ v1	Critical chambers	You	● Active
☐ v2	Daily yield review	You	● Active
☐ v3	PM correlation	Process team	● Shared

3 records

Page Size: 50 1 to 3 of 3 Page 1 of 1

Compact · 38 px

- START
- 

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DataTable Lab
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- 

Support
- 

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- NiceGUI Base 3.0.0a8
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REFERENCE APP PREVIEW · MONITORING

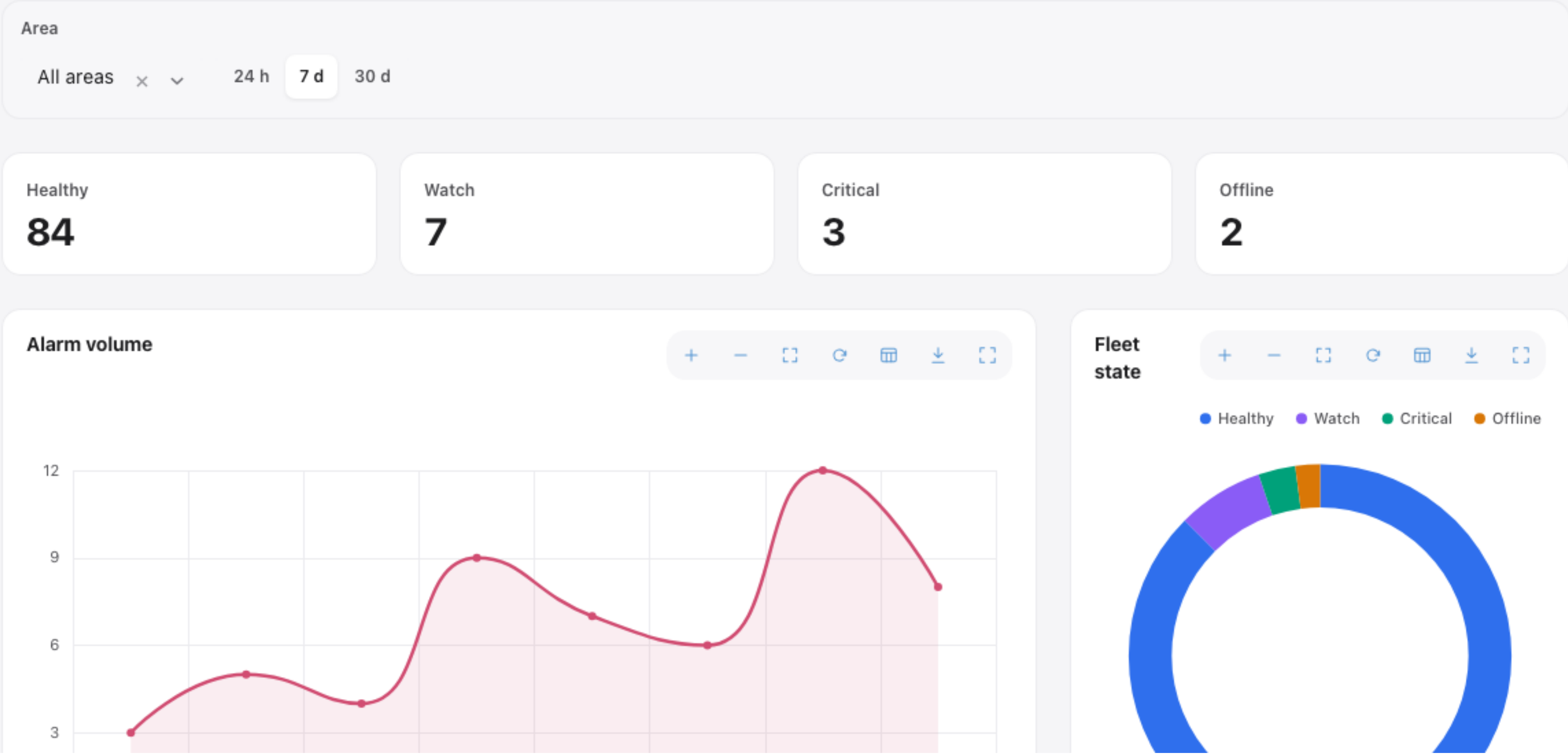
Monitoring

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Fleet Monitoring

Operational health leads; investigation context stays one click away



<

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REFERENCE APP PREVIEW · SEARCH

Search

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Engineering Search

Search-first navigation for tools, chambers, investigations and evidence

Search everything

ETCH-021



Refine

☒ Equipment

☒ Investigations

☒ Evidence

3 results · indexed 2 min ago



ETCH-021

Tool · Watch

3 active excursions · 2 evidence packages



ETCH-021 / CH-3

Chamber · Watch

Strong commonality with EXC-1042



EXC-1042

Investigation · Active

CD drift and edge spatial signature

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NiceGUI 3.15.0

REFERENCE APP PREVIEW · WIZARD

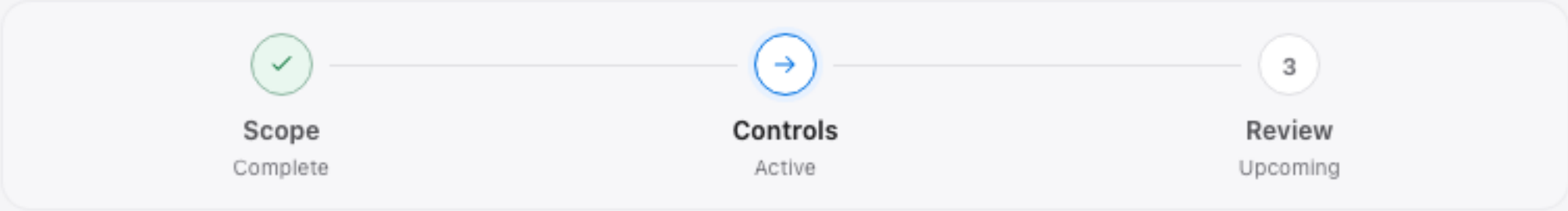
Wizard

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Create Investigation

Guided task with visible progress, bounded decisions and a real review step



Select control population

Use the narrowest defensible comparison population before reviewing evidence.

Control strategy

Exact route match

×

▼

☒ Require exact tool / recipe / chamber sequence

☒ Exclude wafers processed after maintenance

Cancel

> Review controls

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NiceGUI Base / Metrology Enginee...
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REFERENCE APP PREVIEW · COMPARISON

Comparison

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Affected vs Control

Direct comparison keeps population identity, deltas and spatial evidence aligned

Affected population

EXC-1042 · 84 wafers

Control population

Matched route · 215 wafers

Affected mean

42.18 nm

Control mean

39.96 nm

Delta

+2.22 nm

SMD

1.42

AFFECTED



ETCH-021 / CH-3

Affected

84 wafers · excursion population

CONTROL



ETCH-014 / CH-2

Control

215 wafers · matched route

Spatial signature

Affected edge signature vs matched control on one scale

+ - ↺

AFFECTED
97 measured dies



CONTROL
97 measured dies



Shared scale

- START
- 

Overview
- 

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Engineering & RCA

- REFERENCE APPS
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REFERENCE APP PREVIEW · ANALYSIS WORKSPACE

Analysis Workspace

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Excursion Investigation — EXC-1042

Dense analytical workspace with persistent orientation and semiconductor-native evidence

INVESTIGATION	OWNER	STAGE	UPDATED
EXC-1042 · Chamber 3 process degradation	Process Engineering	Evidence review	2 minutes ago

Affected

84 wafers

Controls

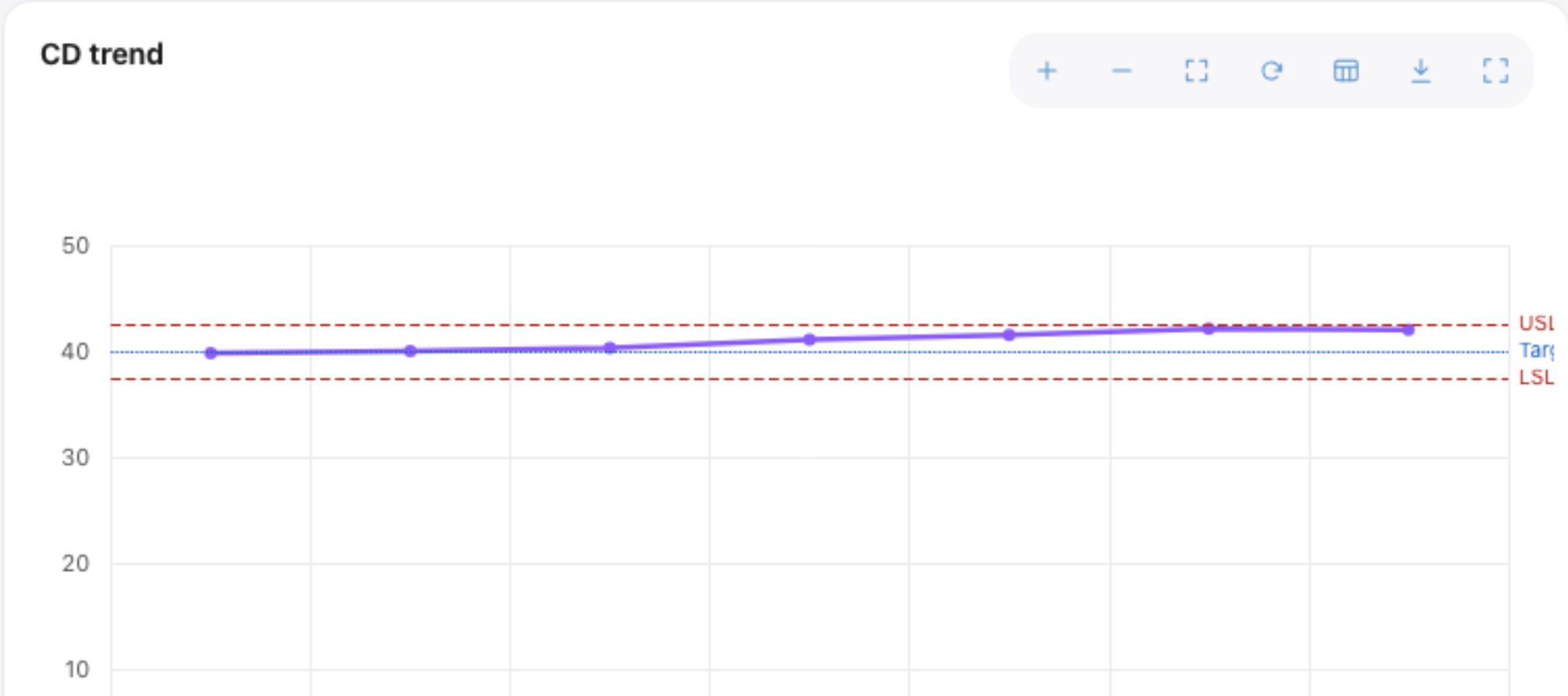
215 wafers

Confidence

High

Contradictions

1





ETCH-021 / CH-3

Watch

Leading equipment commonality

RECIPE

ETCH_R18

LAST PM

17 days

LOTS AFFECTED

12 lots

EVIDENCE CONFIDENCE

High



Working hypothesis

Chamber-specific process drift is supported by CD

<

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · ACTIONS

ButtonGroup

Group related actions without visual fragmentation

BEST FOR

toolbar actions · mode actions

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

- Preview

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ButtonGroup

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Run

Pause

Reset

<

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Theme · System

Appearance

COMPONENT · ACTIONS

SplitButton

Primary action with closely related alternatives

BEST FOR

export alternatives · run options

AVOID WHEN

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› **Advanced contract · configuration, data, accessibility, and authority**

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SplitButton

Interactive specimen

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Export CSV

...

<

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Theme · System

Appearance

COMPONENT · SURFACES

Divider

Subtle structural separation

BEST FOR

sections · menus

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

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PreviewUsageInspectCode

Divider

Interactive specimen

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Measurement summary

<

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · SURFACES

CollapsiblePanel

Progressive disclosure without leaving context

BEST FOR

advanced settings · secondary detail

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

PreviewUsageConfigureStatesInteractionsInspectCode

CollapsiblePanel

Interactive specimen Use the real control below; interactions are recorded in Event / debug output.

Advanced settings

<

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · SURFACES

Accordion

Grouped progressive disclosure

BEST FOR

help · configuration groups

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

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PreviewUsageConfigureStatesInteractionsInspectCode

Accordion

Interactive specimen Use the real control below; interactions are recorded in Event / debug output.

Sampling

Every 10 wafers

<

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · STATUS

Chip

Interactive compact metadata or filter value

BEST FOR

filter value · selection

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Inspect

Code

Chip

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Lot A12

png_sha256=1483c3679ff90e64c85f4ec5306b35d5b3a561272f39aea6483b1fc9e4afa069

<

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Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · STATUS

CountBadge

Compact numeric count

BEST FOR

notifications · selected count

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Inspect

Code

CountBadge

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

24

png_sha256=d42db9909dce25a81900620f6fe1d05d3c893ccde5e63d23fff680792b7fb0c9

<

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · STATUS

SeverityIndicator

Semantic severity with label and non-color cue

BEST FOR

operational state · risk

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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SeverityIndicator

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

● High severity

<

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · STATUS

FreshnessIndicator

Data recency state

BEST FOR

updated time · stale data

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

>

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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Code

FreshnessIndicator

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Updated 2 min ago

<

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · STATUS

DataQualityBadge

Data reliability/completeness state

BEST FOR

partial data · estimated data

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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DataQualityBadge

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Complete

png_sha256=d84c468bdf964028e330bfc7ceaf33a8a8e8909f7f4840fe4957c81d52cc17fb

<

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · ACTIONS

Button

Standard user action

BEST FOR

normal action · secondary action

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

PreviewUsageConfigureStatesInteractionsInspectCode

Button

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

▶ Run analysis

<

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · ACTIONS

ActionButton

Async-safe or stateful action

BEST FOR

save · run analysis

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

PreviewUsageConfigureStatesInteractionsInspectCode

ActionButton

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Save changes

<

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · ACTIONS

IconButton

Compact icon-only action with accessible label

BEST FOR

toolbar action · row action

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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IconButton

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · SURFACES

Surface

Consistent content containment and interaction surface

BEST FOR

panel · card

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

PreviewUsageInspectCode

Surface

Interactive specimen Use the real control below; interactions are recorded in Event / debug output.

Panel

Card

Well

<

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · STATUS

Badge

Compact semantic status or metadata label

BEST FOR

status · severity

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Inspect

Code

Badge

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Healthy

Warning

Excursion

png_sha256=ab935a361fae7a4043b99e9f5fd6f0a817c4f311ab220135f609e3d0e11a0aa7

<

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

TextInput

Single-line text entry

BEST FOR

name · identifier

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

TextInput

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Lot ID

LOT-2409



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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

NumberInput

Numeric entry with bounds and unit

BEST FOR

threshold · count

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

PreviewUsageConfigureStatesInteractionsInspectCode

NumberInput

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Upper limit

<

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K



Theme · System

Appearance

COMPONENT · INPUTS

TextArea

Multi-line text entry

BEST FOR

notes · description

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

TextArea

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Engineering note

Investigate chamber drift.

<

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+KTheme · SystemAppearance

COMPONENT · INPUTS

SearchInput

Debounced search entry

BEST FOR

global search · table search

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

PreviewUsageConfigureStatesInteractionsInspectCode

SearchInput

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Search lots

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

Select

Single choice from known values

BEST FOR

filter · form choice

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

PreviewUsageConfigureStatesInteractionsInspectCode

Select

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Process area

<

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K



Theme · System

Appearance

COMPONENT · INPUTS

MultiSelect

Multiple choices from known values

BEST FOR

multi-filter · assignment

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

MultiSelect

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Process areas

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K



Theme · System

Appearance

COMPONENT · INPUTS

Autocomplete

Searchable known values

BEST FOR

large option set · entity lookup

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

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Interactions

Inspect

Code

Autocomplete

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Tool

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

Combobox

Search/select with optional custom value

BEST FOR

tag-like entry · mixed known/custom choice

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

PreviewUsageConfigureStatesInteractionsInspectCode

Combobox

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Tag

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

Checkbox

Independent boolean selection

BEST FOR

multi-option form · enable choice

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

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Interactions

Inspect

Code

Checkbox

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Include rework lots

png_sha256=2abf46761befe23c09bf44497b15a81b6b7c3fb418aca08d2972cfe31c832219

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

CheckboxGroup

Set of independent boolean choices

BEST FOR

feature selection · permissions

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

>

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

- Preview

Usage

Configure

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Interactions

Inspect

Code

CheckboxGroup

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Signals

Etch

Deposition

png_sha256=fd20275c55b1d46e8a75df276ab6946350fffd24bf715cc07233395e807e26ab

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

RadioGroup

Mutually exclusive choice

BEST FOR

mode selection · single preference

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

RadioGroup

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Mode

- ☐

Etch
- ☐

Deposition

<

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

Switch

Immediate on/off setting

BEST FOR

preference toggle · feature enablement

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

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Interactions

Inspect

Code

Switch

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Auto refresh

<

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

Slider

Bounded continuous or stepped value

BEST FOR

threshold · range tuning

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

States

Interactions

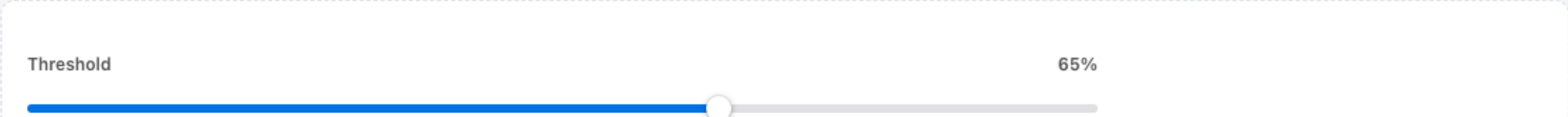
Inspect

Code

Slider

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.



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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K



Theme · System

Appearance

COMPONENT · INPUTS

RangeSlider

Bounded low/high selection

BEST FOR

numeric filtering · window selection

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. [Live example](#) is ready.

Preview

Usage

Configure

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Interactions

Inspect

Code

RangeSlider

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Window

20–80%

<

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REFERENCE

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

DatePicker

Single date selection

BEST FOR

effective date · event date

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

DatePicker

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Effective date

<

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

DateRangePicker

Date interval selection

BEST FOR

analysis period · reporting window

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

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Interactions

Inspect

Code

DateRangePicker

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Analysis period

<

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

TimePicker

Time-of-day selection

BEST FOR

schedule time · cutoff

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

- Preview

Usage

Configure

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Interactions

Inspect

Code

TimePicker

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Cutoff time

<

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

DateTimePicker

Date and time selection

BEST FOR

timestamp · scheduled action

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

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Interactions

Inspect

Code

DateTimePicker

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Scheduled run

<

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K



Theme · System

Appearance

COMPONENT · INPUTS

FileUpload

Validated file selection and upload

BEST FOR

attachment · data import

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

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Inspect

Code

FileUpload

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Upload process data

Click to browse, drag files here, or paste from the clipboard with Ctrl+V / Cmd+V.

0.0B / 0.00%



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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Dashboard

High-level KPI and trend overview.

BEST FOR

High-level KPI and trend overview.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

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Dashboard

At-a-glance process performance with trends and exceptions.

Lots today

128

In control

96.9%

Open exceptions

3

<

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Data Explorer

Interactive filtering, analysis, records and contextual drill-down.

BEST FOR

Interactive filtering, analysis, records and contextual drill-down.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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Inspect

Code

Data Explorer

Filter records, inspect the governed table, and retain selected context.

Search lots

<

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REFERENCE

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K



Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Master Detail

Browse entities while preserving selected-entity context.

BEST FOR

Browse entities while preserving selected-entity context.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Data

Inspect

Code

Master Detail

Browse lots while the selected lot remains visible beside the master list.

Find lot or tool

ETCH

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K

Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Crud

Search, create, inspect and edit managed records.

BEST FOR

Search, create, inspect and edit managed records.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Data

Inspect

Code

Crud

Create, read, update, validate, and safely delete managed records.

Search managed recipes

POLY

Create recipe

Delete selected...

<

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REFERENCE

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Monitoring

Operational health, alerts and periodically refreshed data.

BEST FOR

Operational health, alerts and periodically refreshed data.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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Monitoring

Lead with operational health, trend, alerts, and supporting evidence.

Fleet health

11 / 12

Active alarms

2

Data freshness

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PATTERN · APPLICATION PATTERN

Search

Search and refine heterogeneous or entity-oriented results.

BEST FOR

Search and refine heterogeneous or entity-oriented results.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

PreviewUsageInspectCode

Search

Query and refine entity results without making an empty state the default.

Search lots, tools, or recipes

8 results for "ETCH" · Status: Any status

LOT-240901

<

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Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Settings

Structured application/user configuration.

BEST FOR

Structured application/user configuration.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

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Preview

Usage

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Settings

Navigate settings, edit governed fields, validate, save, and reset local state.

Settings sections

Data refresh

These values apply to this local specimen only.

<

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Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Wizard

Guided multi-step task with clear progress and bounded decisions.

BEST FOR

Guided multi-step task with clear progress and bounded decisions.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Inspect

Code

Wizard

Complete a bounded task with progress, validation, review, Back, and Next.

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Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Comparison

Compare baseline/current populations, entities or scenarios.

BEST FOR

Compare baseline/current populations, entities or scenarios.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Data

Inspect

Code

Comparison

Align two populations and make deltas and evidence immediately comparable.

Baseline

Golden tool · ETCH-03

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K



Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Analysis Workspace

Maximum-density resizable chart/table analysis environment.

BEST FOR

Maximum-density resizable chart/table analysis environment.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

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Analysis Workspace

Keep analysis controls, visualization, records, and selected context together.

Tool

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Theme · System

Appearance

ANALYTIC · SPC

SPC I-MR

Individuals and moving-range control chart

BEST FOR

continuous measurements without rational subgroups

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

SPCMonitor

PMEffectAnalysis

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

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Code

Control chart · Individual + moving range · x: sample order · y: measurement

Individuals + Moving Range · x: sample order / adjacent samples · y: Measurement / absolute difference

Individuals — SPC I-MR

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Theme · System

Appearance

ANALYTIC · SPC

SPC Xbar-R

Xbar-R subgroup control chart

BEST FOR

small equal-size rational subgroups

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

>

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

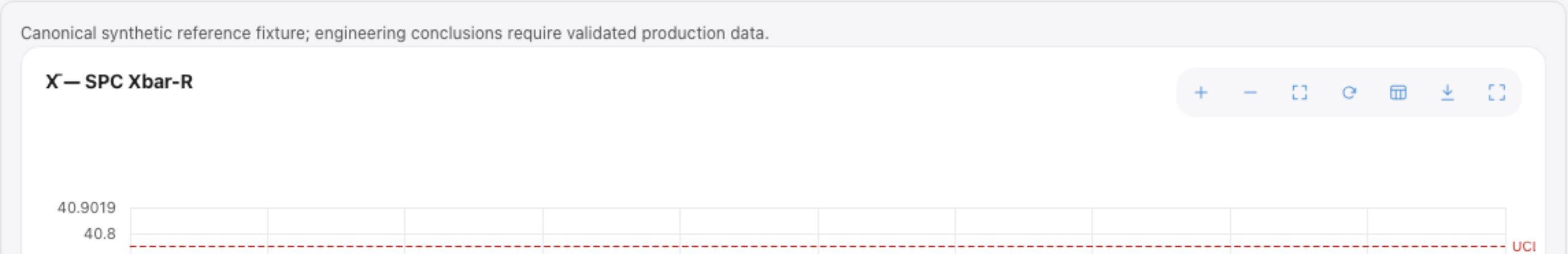
- Preview

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ANALYTIC · SPC

SPC Xbar-S

Xbar-S subgroup control chart

BEST FOR
equal-size subgroup variability matters

AVOID WHEN
when chart assumptions, subgroup structure, or baseline definition do not match the data

> **Advanced contract · configuration, data, accessibility, and authority**

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Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

X̄ — SPC Xbar-S

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ANALYTIC · SPC

SPC P

Proportion nonconforming control chart

BEST FOR

binary defect proportion with varying samples

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

SPC P

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Nonconforming proportion

Center

UCL

LCL



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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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Theme · System

Appearance

ANALYTIC · SPC

SPC Np

Count nonconforming control chart

BEST FOR

binary defect counts with constant sample size

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

>

Advanced contract · configuration, data, accessibility, and authority

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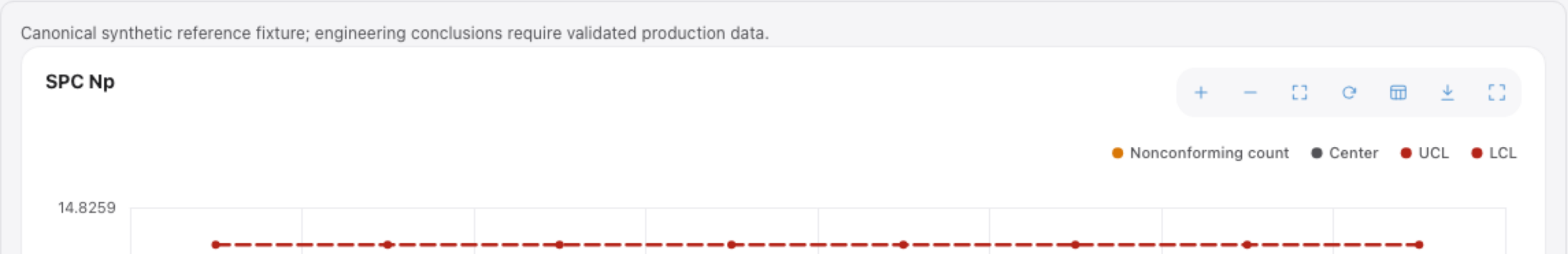
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Appearance

ANALYTIC · SPC

SPC C

Defect count control chart

BEST FOR

Poisson defect counts with constant opportunity

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

> **Advanced contract · configuration, data, accessibility, and authority**

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

SPC C

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Defect count Center UCL LCL



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ANALYTIC · SPC

SPC U

Defects-per-unit control chart

BEST FOR

Poisson defects with varying opportunity

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Data

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

SPC U

+

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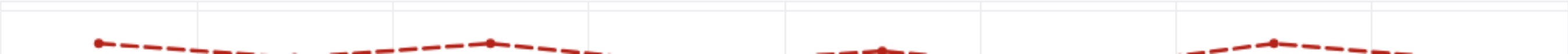
Defects / unit

Center

UCL

LCL

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ANALYTIC · SPC

SPC EWMA

EWMA drift-sensitive control chart

BEST FOR
small sustained process shifts

AVOID WHEN
when chart assumptions, subgroup structure, or baseline definition do not match the data

› **Advanced contract · configuration, data, accessibility, and authority**

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Preview

Usage

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Code

Control chart · EWMA smoothed statistic · x: sample order · y: EWMA value

SPC EWMA

+

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EWMA

Center

UCL

LCL

41.2337

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ANALYTIC · SPC

SPC CUSUM

CUSUM change-detection chart

BEST FOR

small persistent mean shifts

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

> Advanced contract · configuration, data, accessibility, and authority

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- Preview

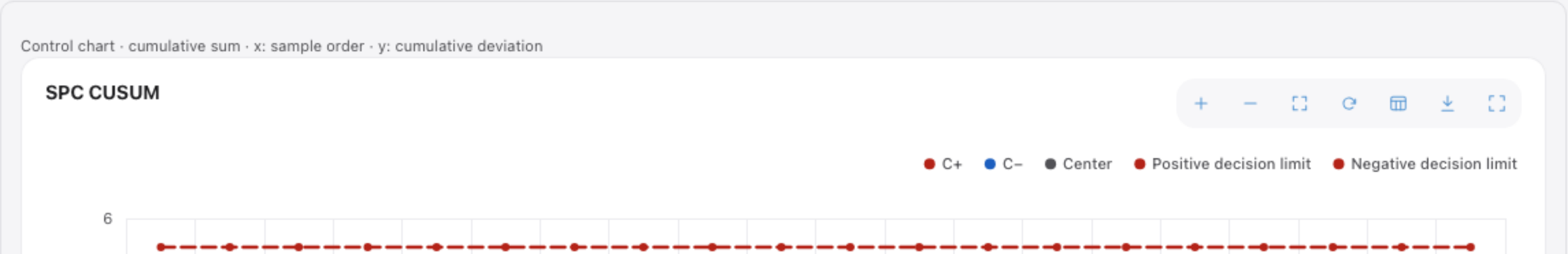
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Theme · System

Appearance

ANALYTIC · CAPABILITY

Capability Histogram

Capability distribution with spec limits

BEST FOR

Cp/Cpk/Pp/Ppk and distribution shape

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

› **Advanced contract · configuration, data, accessibility, and authority**

Related capabilities

SPCMonitor

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Data

Configure

Inspect

Code

Histogram · x: measurement bins · y: Count

Capability Histogram

Numeric measurement bins with specification markers on the same coordinate axis.

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ANALYTIC · CAPABILITY

Q-Q Probability

Normal probability/ QQ diagnostic

BEST FOR

capability normality assumption review

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

> Advanced contract · configuration, data, accessibility, and authority

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Preview

Usage

Data

Configure

Inspect

Code

Q-Q probability plot · x: Theoretical quantile · y: Observed value

Q-Q Probability

Observed points against a fitted location/scale normal reference line.

Observed

Expected reference

png_sha256=fd455c3827f31b603c93a4107abb92c9ec722df6360b88c44e84183e46b94613

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Theme · System

Appearance

ANALYTIC · CAPABILITY

ECDF

Empirical cumulative distribution

BEST FOR

distribution comparison without binning

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

› Advanced contract · configuration, data, accessibility, and authority

Related capabilities

SPCMonitor

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Preview

Usage

Data

Configure

Inspect

Code

Empirical cumulative distribution · x: Value · y: Cumulative probability

ECDF

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ANALYTIC · CAPABILITY

Box Distribution

Box distribution comparison

BEST FOR

compact affected/control distribution summaries

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

PMEffectAnalysis

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Data

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Code

Box distribution · x: population · y: measurement

Box Distribution

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Theme · System

Appearance

ANALYTIC · CAPABILITY

Violin Distribution

Violin distribution comparison

BEST FOR

shape plus spread comparison

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

> Advanced contract · configuration, data, accessibility, and authority

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Preview

Usage

Data

Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Violin distribution · x: population · width: local density

Violin Distribution

Density is estimated from the canonical population measurements.

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Theme · System

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ANALYTIC · CAPABILITY

Ridge Distribution

Ridge distribution comparison

BEST FOR

many tool/chamber distributions

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

› **Advanced contract · configuration, data, accessibility, and authority**

Related capabilities

[ChamberMatching](#)

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Data

Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Ridge distribution · rows: populations · x: value · height: density

Ridge Distribution

Offset density curves keep each chamber distribution readable.

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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Continuous

Continuous wafer map

BEST FOR

continuous spatial metrology or process response

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

› **Advanced contract · configuration, data, accessibility, and authority**

Related capabilities

SPCMonitor

LotWaferExplorer

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Data

Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Continuous

Die-level wafer signature · wheel/drag to inspect spatial structure

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Theme · System

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ANALYTIC · WAFER

Wafer Categorical

Categorical/bin wafer map

BEST FOR

yield bins or categorical spatial state

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

LotWaferExplorer

YieldLoss

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Preview

Usage

Data

Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Categorical wafer map · legend: Category · die position: x/y

Wafer Categorical

Die-level wafer signature · wheel/drag to inspect spatial structure

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ANALYTIC · WAFER

Wafer Defect

Defect wafer map

BEST FOR
inspection defects and defect classes

AVOID WHEN
when coordinates are unavailable or spatial sampling is too sparse for the intended inference

› **Advanced contract · configuration, data, accessibility, and authority**

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Preview

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Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Defect wafer map · legend: Defect state · die position: x/y

Wafer Defect

Die-level wafer signature · wheel/drag to inspect spatial structure

CENTER ↔ EDGE SIGNATURE

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Search ⌘K / Ctrl+K

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ANALYTIC · WAFER

Wafer Delta

Affected minus control wafer delta map

BEST FOR

spatial process change localization

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ExcursionDefenseLine

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Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Delta

Die-level wafer signature · wheel/drag to inspect spatial structure

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Search

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ANALYTIC · WAFER

Wafer Comparison

Side-by-side wafer comparison

BEST FOR

affected/control wafer signature comparison

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

› **Advanced contract · configuration, data, accessibility, and authority**

Related capabilities

ChamberMatching

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Comparison

Synchronized affected vs control wafer signature · one shared scale



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ANALYTIC · WAFER

Lot Wafer Strip

Lot wafer strip

BEST FOR

lot-level wafer-to-wafer spatial variation

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

[LotWaferExplorer](#)

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

Lot Wafer Strip

Wafer W01

Die-level wafer

Wafer W02

Die-level wafer

Wafer W03

Die-level wafer

Wafer W04

Die-level wafer

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Theme · System

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ANALYTIC · WAFER

Wafer Small Multiples

Synchronized wafer small multiples

BEST FOR

many wafers on one scale

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

› **Advanced contract · configuration, data, accessibility, and authority**

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Small Multiples

Wafer W01

Die-level wafer signature · wheel/drag to inspect spatial structure

Wafer W02

Die-level wafer signature · wheel/drag to inspect spatial structure

Wafer W03

Die-level wafer signature · wheel/drag to inspect spatial structure

Wafer W04

Die-level wafer signature · wheel/drag to inspect spatial structure

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Appearance

ANALYTIC · WAFER

Wafer Contour

Contour/surface wafer map

BEST FOR

data-derived spatial field interpretation

AVOID WHEN

sparse categorical data

Advanced contract · configuration, data, accessibility, and authority

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PreviewUsageDataInspectCode

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Contour

Data-derived continuous wafer field · isolines are clipped to governed wafer geometry

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CLIPPED DATA-DERIVED CONTOUR ISOLINES

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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Radial

Radial profile

BEST FOR

center-edge signatures

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

LotWaferExplorer

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Preview

Usage

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Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Radial

Center → edge profile · compare radial signature without Cartesian chart chrome

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Reference Explorer

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 Theme · System

Appearance

ANALYTIC • WAFER

Wafer Sector

Sector analysis

BEST FOR
azimuthal spatial signatures

AVOID WHEN
when coordinates are unavailable or spatial sampling is too sparse for the intended inference

- Advanced contract · configuration, data, accessibility, and authority

Related capabilities

LotWaferExplorer

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Preview Usage Data Inspect Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Sector

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Theme · System

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ANALYTIC · WAFER

Wafer Defect Clusters

Defect-cluster overlays

BEST FOR

identify spatially clustered inspection defects

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

LotWaferExplorer

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PreviewUsageDataInspectCode

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Defect Clusters

Die-level wafer signature · wheel/drag to inspect spatial structure

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Theme · System

Appearance

ANALYTIC · FDC

FDC Recipe Step Trace

Recipe-step-aligned trace

BEST FOR

compare traces with step semantics

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Recipe Step Trace

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Theme · System

Appearance

ANALYTIC · FDC

FDC Golden Envelope

Golden trace envelope

BEST FOR

compare trace deviation against baseline population

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

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Preview

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Golden Envelope

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Golden upper

Observed

Golden lower

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Theme · System

Appearance

ANALYTIC · FDC

FDC Multi Sensor

Multi-sensor trace panel

BEST FOR

correlated equipment sensor behavior

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

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Appearance

ANALYTIC · FDC

FDC Tool Chamber Compare

Tool/chamber comparison

BEST FOR

matching and PM-effect comparisons

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

PMEffectAnalysis

ChamberMatching

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Tool Chamber Compare

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ANALYTIC · FDC

FDC Chamber Fingerprint

Chamber fingerprint

BEST FOR

multivariate chamber matching

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

PMEffectAnalysis

ChamberMatching

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Preview

Usage

Data

Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Chamber Fingerprint

Normalized chamber/process fingerprint · hover cells for exact contribution

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ANALYTIC · FDC

FDC Sensor Fingerprint

Sensor fingerprint

BEST FOR

sensor signature comparison

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Sensor Fingerprint

Normalized chamber/process fingerprint · hover cells for exact contribution

Mean shift

Slope

Noise

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Theme · System

Appearance

ANALYTIC · FDC

FDC Alarm Overlay

Alarm overlay

BEST FOR

align alarms with trace excursions

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

>

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

PreviewUsageDataInspectCode

Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Alarm Overlay

+ −

2.5

Pressure high

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Theme · System

Appearance

ANALYTIC · FDC

FDC Equipment Event Overlay

Equipment event overlay

BEST FOR

align PM/recipe/equipment events with traces

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

PMEffectAnalysis

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Equipment Event Overlay

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ANALYTIC · FDC

FDC PCA Scores

PCA scores

BEST FOR

multivariate operating-state separation

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

ChamberMatching

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC PCA Scores

+ −

Baseline

Affected

<

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ANALYTIC · FDC

FDC PCA Loadings

PCA loadings

BEST FOR

variables driving multivariate separation

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

ChamberMatching

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FDC PCA Loadings

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ANALYTIC · FDC

FDC Hotelling T²

Hotelling T²

BEST FOR

multivariate score-space anomaly magnitude

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

> **Advanced contract · configuration, data, accessibility, and authority**

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Preview

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Hotelling T²

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Theme · System

Appearance

ANALYTIC · FDC

FDC SPE Q

SPE/Q statistic

BEST FOR

PCA residual anomaly magnitude

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

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FDC SPE Q

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Theme · System

Appearance

ANALYTIC · RCA

RCA Affected Control

Affected/control population comparison

BEST FOR

quantify excursion shift without causal overclaim

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ExcursionDefenseLine

PMEffectAnalysis

RcaCockpit

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Affected Control

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Department standard · Reference Explorer

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Reference Explorer

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Theme · System

Appearance

ANALYTIC · RCA

RCA Commonality Ranking

Commonality ranking

BEST FOR

rank shared factors across affected versus control

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ExcursionDefenseLine

YieldLoss

RcaCockpit

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Live example is ready.

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Commonality Ranking

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Theme · System

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ANALYTIC · RCA

RCA Enrichment

Enrichment analysis

BEST FOR

factor exposure over-representation

AVOID WHEN

as proof of causality without corroborating engineering evidence

> **Advanced contract · configuration, data, accessibility, and authority**

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Preview

Usage

Data

Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Enrichment

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5

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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Theme · System

Appearance

ANALYTIC · RCA

RCA Commonality Matrix

Commonality matrix

BEST FOR

scan factor/value enrichment

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ExcursionDefenseLine

RcaCockpit

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Preview

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Commonality Matrix

RCA factor commonality · darker cells indicate stronger population enrichment

REFERENCE

Member Member

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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Theme · System

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ANALYTIC · RCA

RCA Contribution Waterfall

Contribution waterfall

BEST FOR

show ranked contribution accumulation

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

RcaCockpit

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Live example is ready.

Preview

Usage

Data

Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Contribution Waterfall

Signed bars span each contribution from its prior cumulative total to the reconciled net shift.

Cumulative delta

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ANALYTIC · RCA

RCA Correlation Matrix

Correlation matrix

BEST FOR

numeric relationship screening

AVOID WHEN

causal conclusions

> Advanced contract · configuration, data, accessibility, and authority

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Preview

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Correlation Matrix

Temperature

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ANALYTIC · RCA

RCA Evidence Matrix

Evidence matrix

BEST FOR

hypothesis evidence/contradiction review

AVOID WHEN

as proof of causality without corroborating engineering evidence

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ExcursionDefenseLine

RcaCockpit

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Evidence Matrix

RCA factor commonality · darker cells indicate stronger population enrichment

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ANALYTIC · RCA

RCA Genealogy Graph

Genealogy/process graph

BEST FOR

lot/wafer/tool/recipe route genealogy

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ExcursionDefenseLine

RcaCockpit

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Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Genealogy Graph

Branching and merging entity relationships preserve route order; they do not imply causality.

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ANALYTIC · RCA

RCA Cause Tree

Cause tree

BEST FOR

structured causal candidates

AVOID WHEN

as proof of causality without corroborating engineering evidence

> Advanced contract · configuration, data, accessibility, and authority

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Cause Tree

Hierarchical candidate decomposition; candidates remain hypotheses until evidence corroborates them.

CH-3 drift

<

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Reference Explorer

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ANALYTIC · RCA

RCA Fault Tree

Fault tree

BEST FOR

logical failure decomposition

AVOID WHEN

as proof of causality without corroborating engineering evidence

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

RcaCockpit

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Fault Tree

AND/OR logic is explicit and remains separate from evidence confidence.

<

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ANALYTIC · RCA

RCA Sankey

Sankey/process flow

BEST FOR

population flow through route/tool/process paths

AVOID WHEN

as proof of causality without corroborating engineering evidence

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

[RcaCockpit](#)

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Sankey

Band width encodes wafer quantity through route, tool, chamber, and review stages.

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ANALYTIC · YIELD

Yield Pareto

Yield loss Pareto

BEST FOR

rank yield-loss contributors

AVOID WHEN

when numerator/denominator or bin definitions are inconsistent across populations

> **Advanced contract · configuration, data, accessibility, and authority**

Related capabilities

YieldLoss

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

Yield Pareto

+

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Contributors

Cumulative % Cumulative

<

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ANALYTIC · YIELD

Bin Pareto

Bin Pareto

BEST FOR
rank failing bins

AVOID WHEN
when numerator/denominator or bin definitions are inconsistent across populations

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

YieldLoss

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Bin Pareto

t / cumulative percent

Contributors

Cumulative %

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ANALYTIC · YIELD

Yield Waterfall

Yield waterfall/decomposition

BEST FOR

explain yield delta composition

AVOID WHEN

when numerator/denominator or bin definitions are inconsistent across populations

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

YieldLoss

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PreviewUsageDataInspectCode

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Yield Waterfall

Loss and recovery bars reconcile the signed total yield change.

Cumulative delta

<

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Appearance

ANALYTIC · RELIABILITY

Weibull Reliability

Weibull reliability analysis

BEST FOR

time-to-failure reliability data

AVOID WHEN

when failure/censoring semantics or exposure time are unknown

› **Advanced contract · configuration, data, accessibility, and authority**

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Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Weibull Reliability

β 1.35 · η 91.8 · R^2 0.96 · failures 9 · censored 3



● Failures

● Censored

● Weibull fit

1

0.9

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ANALYTIC · DOE

DOE Main Effects

DOE main effects

BEST FOR

screen factor effects

AVOID WHEN

for observational data that were not generated by a valid designed experiment

[Advanced contract · configuration, data, accessibility, and authority](#)

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Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

DOE Main Effects

+

−

RF bias

Pressure

41.4



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ANALYTIC · DOE

DOE Interactions

DOE interaction plot

BEST FOR
factor interaction assessment

AVOID WHEN
for observational data that were not generated by a valid designed experiment

› **Advanced contract · configuration, data, accessibility, and authority**

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Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

DOE Interactions

+

−

● Pressure low

● Pressure high

41.3

A1

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Theme · System

Appearance

ANALYTIC · DOE

DOE Response Surface

DOE response surface

BEST FOR

quadratic two-factor response optimization

AVOID WHEN

for observational data that were not generated by a valid designed experiment

> Advanced contract · configuration, data, accessibility, and authority

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RECIPE · SEMICONDUCTOR RECIPE

SPCMonitor

Monitor process stability, capability and violations with drill-down to manufacturing context.

BEST FOR

monitor a process parameter · detect process drift or control-rule violations

AVOID WHEN

when the process has no defensible baseline or chart family/subgroup definition

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

SPC I-MR

Capability Histogram

ECDF

Wafer Continuous

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SPCMonitor

Canonical sample composition

Confirm a Data mapping to restrict this composition to panels supported by your own development dataset.

ANALYTICAL · SPC_I_MR

ANALYTICAL · CAPABILITY_HISTOGRAM

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Theme · System

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RECIPE · SEMICONDUCTOR RECIPE

ExcursionDefenseLine

Triage an excursion with affected/control populations, spatial localization, commonality and genealogy evidence.

BEST FOR

build an excursion defense line · triage affected versus control material

AVOID WHEN

when affected/control population definitions are not yet governed or comparable

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

RCA Affected Control

Wafer Delta

RCA Commonality Ranking

RCA Commonality Matrix

RCA Genealogy Graph

RCA Evidence Matrix

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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ExcursionDefenseLine

Canonical sample composition

Confirm a Data mapping to restrict this composition to panels supported by your own development dataset.

ANALYTICAL · RCA_AFFECTED_CONTROL

ANALYTICAL · WAFER_DELTA

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Theme · System

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RECIPE · SEMICONDUCTOR RECIPE

FdcToolHealth

Diagnose equipment health from aligned sensor traces, envelopes, events and multivariate fingerprints.

BEST FOR

analyze FDC traces · check tool or chamber health

AVOID WHEN

when recipe-step alignment or sensor/event semantics are unavailable

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FDC Recipe Step Trace

FDC Golden Envelope

FDC Multi Sensor

FDC Chamber Fingerprint

FDC PCA Scores

FDC PCA Loadings

FDC Equipment Event Overlay

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FdcToolHealth

Canonical sample composition

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ANALYTICAL · FDC_RECIPE_STEP_TRACE

ANALYTICAL · FDC_GOLDEN_ENVELOPE

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Theme · System

Appearance

RECIPE · SEMICONDUCTOR RECIPE

LotWaferExplorer

Explore lot-to-wafer-to-die spatial behavior with synchronized maps, defects, profiles and records.

BEST FOR

explore wafers in a lot · inspect spatial metrology or defect signatures

AVOID WHEN

when wafer/die coordinates or wafer identity are unavailable

› Advanced contract · configuration, data, accessibility, and authority

Related capabilities

- [Wafer Continuous](#)
- [Wafer Categorical](#)
- [Lot Wafer Strip](#)
- [Wafer Radial](#)
- [Wafer Ring](#)
- [Wafer Sector](#)
- [Wafer Defect Clusters](#)

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LotWaferExplorer

Canonical sample composition

Confirm a Data mapping to restrict this composition to panels supported by your own development dataset.

ANALYTICAL · WAFER_CONTINUOUS

ANALYTICAL · WAFER_CATEGORICAL

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Theme · System

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RECIPE · SEMICONDUCTOR RECIPE

YieldLoss

Explain yield loss through Pareto/decomposition, bin/spatial signatures and commonality drill-down.

BEST FOR

analyze yield loss · rank fail bins or loss contributors

AVOID WHEN

when yield denominators, bin definitions or comparison populations are inconsistent

› **Advanced contract · configuration, data, accessibility, and authority**

Related capabilities

[Yield Pareto](#)

[Bin Pareto](#)

[Yield Waterfall](#)

[Wafer Categorical](#)

[RCA Commonality Ranking](#)

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YieldLoss

Canonical sample composition

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ANALYTICAL · YIELD_PARETO

ANALYTICAL · BIN_PARETO

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RECIPE · SEMICONDUCTOR RECIPE

PMEffectAnalysis

Compare pre/post-PM populations, chamber behavior, traces and process stability using one shared context.

BEST FOR

measure PM effect · compare chamber before and after maintenance

AVOID WHEN

when pre/post windows or maintenance-event timing cannot be established

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

RCA Affected Control

FDC Tool Chamber Compare

SPC I-MR

Box Distribution

FDC Equipment Event Overlay

FDC Chamber Fingerprint

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PMEffectAnalysis

Canonical sample composition

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ANALYTICAL · RCA_AFFECTED_CONTROL

ANALYTICAL · FDC_TOOL_CHAMBER_COMPARE

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RECIPE · SEMICONDUCTOR RECIPE

ChamberMatching

Compare chamber signatures and distributions to detect mismatch and identify driving variables.

BEST FOR

match chambers · compare chambers across a toolset

AVOID WHEN

when chambers do not share comparable process/recipe populations

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

- FDC Chamber Fingerprint

FDC Tool Chamber Compare

FDC PCA Scores

FDC PCA Loadings

Ridge Distribution

Wafer Comparison

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PreviewUsageDataInteractionsInspectCode

ChamberMatching Canonical sample composition

Confirm a Data mapping to restrict this composition to panels supported by your own development dataset.

ANALYTICAL · FDC_CHAMBER_FINGERPRINT

ANALYTICAL · FDC_TOOL_CHAMBER_COMPARE

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RECIPE · SEMICONDUCTOR RECIPE

RcaCockpit

Run structured root-cause analysis with populations, commonality, contribution, evidence and genealogy in one cockpit.

BEST FOR

perform RCA · investigate root cause or commonality

AVOID WHEN

when the workflow needs causal proof from observational association alone

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

RCA Affected Control

RCA Commonality Ranking

RCA Commonality Matrix

RCA Contribution Waterfall

RCA Evidence Matrix

RCA Genealogy Graph

RCA Fault Tree

RCA Sankey

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RcaCockpit

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REFERENCE

Full Applications

Open a complete reference composition built from the department pattern and recipe authorities.

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⌘K / Ctrl+K



Theme · System

Appearance

SPC Control Center

A complete stability review composed from monitoring, dashboard, comparison and drill-down surfaces.

Is the critical-dimension process stable, and which observations need review?

critical dimension · control limits · review queue

Search records

Filter this application example

View

All records



critical dimension

control limits

review queue

Records

12

Synthetic reference fixture

Review focus

critical dimension

Domain context

Pattern

Monitoring

Primary analytical view

Chart and supporting records stay linked to the same filtered fixture.

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REFERENCE

Full Applications

Open a complete reference composition built from the department pattern and recipe authorities.

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Theme · System

Appearance

FDC Tool Health Center

A full operational surface for aligned traces, chamber context, event review and affected records.

Which tool or chamber signal changed around the observed equipment event? pressure trace · tool/chamber · event review

Search records

Filter this application example

View

All records

pressure trace

tool/chamber

event review

Records

12

Synthetic reference fixture

Review focus

pressure trace

Domain context

Pattern

Monitoring

Primary analytical view

Chart and supporting records stay linked to the same filtered fixture.

Full Applications

Open a complete reference composition built from the department pattern and recipe authorities.

Search

⌘K / Ctrl+K



Theme · System

Appearance

Excursion Investigation

An evidence-first investigation workspace for affected/control separation, commonality and supporting records.

What affected/control differences are worth investigating for this excursion? affected vs control · evidence ledger · hypothesis review

Search records

Filter this application example

View

All records



affected vs control

evidence ledger

hypothesis review

Primary analytical view

Chart and supporting records stay linked to the same filtered fixture.

Records

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Synthetic reference fixture

Review focus

affected vs control

Domain context

Pattern

Analysis Workspace

Open selected detail

EVIDENCE

CAVEATS